

General Relativity BigHW-2

PROBLEM 1. Generalize a Schwarzschild black hole solution to a Schwarzschild-de Sitter solution. That is, a spherically-symmetric solution which in the asymptotic spatial infinity goes to a de Sitter space-time. This solution can be obtained by solving equations of motion which follow from the following action

$$S = \int d^4x \sqrt{-g} \left(\frac{M_P^2}{2} R - \Lambda \right)$$

where Λ is a positive constant. What happens if $\Lambda < 0$?

SOLUTION:

By variation we know that EoM is

$$\frac{1}{8\pi G} G_{\mu\nu} = -\frac{1}{2} \Lambda g_{\mu\nu} \implies R_{\mu\nu} = \left(\frac{R}{2} - 4\pi G \Lambda \right) g_{\mu\nu}$$

In spherically symmetric case, we still use coordinates (t, r, Ω) , and we consider the same metric as we have considered in Schwarzschild black hole

$$ds^2 = -e^{2\alpha} dt^2 + e^{2\beta} dr^2 + r^2 d\Omega^2$$

and the components of Ricci tensor are (as we have derived before)

$$R_{00} = [\partial_0^2 \beta + (\partial_0 \beta)^2 - \partial_0 \alpha \partial_0 \beta] + e^{2(\alpha-\beta)} [\partial_1^2 \alpha + (\partial_1 \alpha)^2 - \partial_1 \alpha \partial_1 \beta + \frac{2}{r} \partial_1 \alpha]$$

$$R_{11} = -[\partial_1^2 \alpha + (\partial_1 \alpha)^2 - \partial_1 \alpha \partial_1 \beta - \frac{2}{r} \partial_1 \beta] + e^{2(\beta-\alpha)} [\partial_0^2 \beta + (\partial_0 \beta)^2 - \partial_0 \alpha \partial_0 \beta]$$

$$R_{01} = \frac{2}{r} \partial_0 \beta$$

$$R_{22} = e^{-2\beta} [r(\partial_1 \beta - \partial_1 \alpha) - 1] + 1$$

$$R_{33} = R_{22} \sin^2 \theta$$

Identically we have $R_{01} = 0$ because $g_{\mu\nu}$ is diagonal, thus $\partial_0 \beta = 0$. Let $\alpha = \alpha(r)$, then

$$R = -e^{-2\beta} [\partial_1^2 \alpha + (\partial_1 \alpha)^2 - \partial_1 \alpha \partial_1 \beta + \frac{2}{r} \partial_1 \alpha] - e^{-2\beta} [\partial_1^2 \alpha + (\partial_1 \alpha)^2 - \partial_1 \alpha \partial_1 \beta - \frac{2}{r} \partial_1 \beta]$$

$$+ \frac{2}{r^2} e^{-2\beta} [r(\partial_1 \beta - \partial_1 \alpha) - 1] + \frac{2}{r^2} = 2e^{-2\beta} [-\partial_1^2 \alpha + \partial_1 \alpha \partial_1 \beta + \frac{2}{r} (\partial_1 \beta - \partial_1 \alpha) - \frac{1}{r} - (\partial_1 \alpha)^2] + \frac{2}{r^2}$$

$$R_{00} = \left(\frac{R}{2} - 4\pi G \Lambda \right) g_{00} \implies$$

$$\partial_1^2 \alpha + (\partial_1 \alpha)^2 - \partial_1 \alpha \partial_1 \beta + \frac{2}{r} \partial_1 \alpha = -(-\partial_1^2 \alpha + \partial_1 \alpha \partial_1 \beta + \frac{2}{r} (\partial_1 \beta - \partial_1 \alpha) - \frac{1}{r^2} - (\partial_1 \alpha)^2 - 4\pi G \Lambda e^{2\beta} + \frac{1}{r^2} e^{2\beta})$$

$$\implies -\frac{2}{r}\partial_1\beta + \frac{1}{r^2} + 4\pi G\Lambda e^{2\beta} - \frac{e^{2\beta}}{r^2} = 0$$

Let $h(r) = re^{-2\beta(r)}$, then

$$\frac{h' - \frac{h}{r}}{hr} + \frac{1}{r^2} + 4\pi G\Lambda \frac{r}{h} - \frac{1}{hr} = 0 \implies h' + 4\pi G\Lambda r^2 - 1 = 0 \implies h = C - \frac{4\pi G\Lambda}{3}r^3 + r$$

Thus $e^{2\beta} = \frac{r}{C - \frac{4\pi G\Lambda}{3}r^3 + r}$. In order to return to Schwarzschild metric with limit $\Lambda \rightarrow 0$, we must have $C = -2GM$.

Similarly to solve α , we expand

$$R_{11} = \left(\frac{R_{\mu\nu}g^{\mu\nu}}{2} - 4\pi G \right) g_{11} \implies$$

$$\begin{aligned} \partial_1^2\alpha + (\partial_1\alpha)^2 - \partial_1\alpha\partial_1\beta - \frac{2}{r}\partial_1\beta &= -(-\partial_1^2\alpha + \partial_1\alpha\partial_1\beta + \frac{2}{r}(\partial_1\beta - \partial_1\alpha) - \frac{1}{r^2} - (\partial_1\alpha)^2 - 4\pi G\Lambda e^{2\beta} + \frac{1}{r^2}e^{2\beta}) \\ \implies \frac{2}{r}\partial_1\alpha + \frac{1}{r^2} + 4\pi G\Lambda e^{2\beta} - \frac{e^{2\beta}}{r^2} &= 0 \implies \frac{2}{r}\partial_1\alpha = -\frac{2}{r}\partial_1\beta \\ \implies \alpha = -\beta + C \implies e^{2\alpha} &= Ce^{-2\beta} = C \frac{-2GM + r - \frac{4\pi G\Lambda}{3}r^3}{r} \end{aligned}$$

In order to recover Schwarzschild black hole with $\Lambda \rightarrow 0$ limit, we must have $C = 1$.

Thus the solution is

$$ds^2 = -\frac{-2GM + r - \frac{4\pi G\Lambda}{3}r^3}{r} dt^2 + \frac{r}{-2GM - \frac{4\pi G\Lambda}{3}r^3 + r} dr^2 + r^2 d\Omega^2$$

Now we examine the root of $f(r) = -\frac{4\pi G\Lambda}{3}r^3 + r - 2GM$. When $\Lambda > 0$, since apparently $f(0) < 0$ and $\lim_{r \rightarrow -\infty} f(r) = +\infty$, there must be a root $r_1 < 0$. $f'(r) = 1 - 4\pi G\Lambda r^2$. Let $r_0 = \sqrt{\frac{1}{4\pi G\Lambda}}$, then

$$f(r_0) = -2GM + \frac{2}{3}\sqrt{\frac{1}{4\pi G\Lambda}}$$

When $\frac{1}{9\pi G\Lambda} > 4GM$, there exist two positive roots $r_2 < r_3$, where probably r_2 is the black hole horizon and r_3 is the cosmic horizon?

For $\Lambda > 0, f'(r) > 0, \forall r > 0, \lim_{r \rightarrow +\infty} f(r) = +\infty$, thus there exists uniquely one root $r_1 > 0$.

PROBLEM 2. Prove that Kerr metric is indeed a solution to Einstein equations.

SOLUTION:

We know that the Kerr metric is

$$ds^2 = -dt^2 + \frac{r^2 + a^2 \cos^2 \theta}{r^2 - 2GMr + a^2} dr^2 + (r^2 + a^2 \cos^2 \theta) d\theta^2 + (r^2 + a^2) \sin^2 \theta d\phi^2 + \frac{2GMr}{r^2 + a^2 \cos^2 \theta} (a \sin^2 \theta d\phi - dt)^2$$

Consider the inverse coordinate transformation of

$$x = \sqrt{r^2 + a^2} \sin \theta \cos \phi, y = \sqrt{r^2 + a^2} \sin \theta \sin \phi, z = r \cos \theta$$

then

$$\begin{aligned} ds^2 = -dt^2 + dx^2 + dy^2 + dz^2 + (r^2 + a^2 \cos^2 \theta) &\left(\frac{1}{r^2 - 2GMr + a^2} - \frac{1}{r^2 + a^2} \right) \left(\frac{xdx + ydy + (1 + \frac{a^2}{r^2})dz}{r(r^2 + a^2 \cos^2 \theta)} \right)^2 \\ &+ \frac{2GMr}{r^2 + a^2 \cos^2 \theta} \left(dt - \frac{xdy - ydx}{r^2 + a^2} \right)^2 \end{aligned}$$

$$= -dt^2 + dx^2 + dy^2 + dz^2 + \frac{2GMr}{(r^2 + a^2 \cos^2 \theta) r^2 (r^2 + a^2) (r^2 - 2GMr + a^2)}$$

or

$$\text{Let } \rho = r^2 + a^2 \cos^2 \theta, \Delta = r^2 - 2GMr + a^2.$$

$$\begin{aligned} \Gamma_{00}^1 &= \frac{2GM\Delta(r^2 - a^2 \cos^2 \theta)}{2\rho^3}, \Gamma_{00}^2 = -\frac{2GMa^2 r \sin \theta \cos \theta}{\rho^3}, \Gamma_{01}^0 = \frac{2GM(r^2 + a^2)(r^2 - a^2 \cos^2 \theta)}{2\rho^2 \Delta} \\ \Gamma_{01}^3 &= \frac{2GM(r^2 - a^2 \cos^2 \theta)}{2\rho^2 \Delta}, \Gamma_{02}^0 = -\frac{2GMa^2 r \sin \theta \cos \theta}{\rho^2}, \Gamma_{02}^3 = -\frac{2GMa r \cot \theta}{\rho^2} \\ \Gamma_{03}^1 &= -\frac{\Delta 2GMa \sin^2 \theta (r^2 - a^2 \sin^2 \theta)}{2\rho^3}, \Gamma_{03}^2 = \frac{2GMa r (r^2 + a^2) \sin \theta \cos \theta}{\rho^3}, \Gamma_{11}^1 = \frac{2ra^2 \sin^2 \theta}{2\rho \Delta} \\ \Gamma_{11}^2 &= \frac{a^2 \sin \theta \cos \theta}{\rho \Delta}, \Gamma_{12}^1 = -\frac{a^2 \sin \theta \cos \theta}{\rho}, \Gamma_{12}^2 = \frac{r}{\rho}, \Gamma_{22}^1 = -\frac{r \Delta}{\rho} \\ \Gamma_{22}^2 &= -\frac{a^2 \sin \theta \cos \theta}{\rho}, \Gamma_{23}^3 = \frac{\cot \theta}{\rho^2} [\rho^2 + 2GMa^2 r \sin^2 \theta], \Gamma_{23}^0 = \frac{2GMa^3 r \sin^3 \theta \cos \theta}{\rho^2} \\ &\quad \Gamma_{13}^0, \Gamma_{13}^3, \Gamma_{33}^1, \Gamma_{33}^2 \end{aligned}$$

Then we can try to calculate Ricci tensor:

$$R_{\mu\nu} = \partial_\rho \Gamma_{\mu\nu}^\rho - \partial_\nu \Gamma_{\rho\mu}^\rho + \Gamma_{\nu\lambda}^\rho \Gamma_{\rho\mu}^\lambda - \Gamma_{\rho\lambda}^\rho \Gamma_{\nu\mu}^\lambda$$

$$R_{00} = \partial_1 \Gamma_{00}^1 + \partial_2 \Gamma_{00}^2 + \Gamma_{20}^3 \Gamma_{30}^2 + \Gamma_{10}^3 \Gamma_{30}^1 + \Gamma_{10}^0 \Gamma_{00}^1 + \Gamma_{20}^0 \Gamma_{00}^2 - \Gamma_{00}^1 (\Gamma_{01}^0 + \Gamma_{11}^1 + \Gamma_{21}^2 + \Gamma_{31}^3)$$

I don't think it can be actually calculated by hand...(not in a limited time),so please forgive me that I don't have enough time to calculate it,so check my code BigHw2-2.nb where I verify that the Ricci tensor of Kerr metric is trivial,and it automatically satisfies Einstein equations.